Exploratory Data Analysis on titanic dataset

1. Import necessary libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

2. Read external csv file and store as dataframe.

```
df = pd.read_csv("titanic.csv")
```

3. Return first 5 rows from dataframe.

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

4. Remove unnecessary columns that aren't useful for EDA.

```
df.drop(['PassengerId', 'Name', 'Ticket', 'Fare', 'Cabin',
'Embarked'], axis=1, inplace=True])
```

```
df.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch
0	0	3	male	22.0	1	0
1	1	1	female	38.0	1	0
2	1	3	female	26.0	0	0
3	1	1	female	35.0	1	0
4	0	3	male	35.0	0	0

Insights:

- Survived, Sex – binary categorical variable.
- Pclass – Ordinal categorical variable.
- Age – Continuous numerical variable.
- SibSp, Parch – Discrete numerical variable.

5. Check for null values; if they are present in significant amount, impute them with statistical value or if they are in small count, simply drop them from dataframe.

```
df.isnull().sum()
```

```
Survived      0
Pclass        0
Sex           0
Age         177
SibSp         0
Parch         0
dtype: int64
```

Age column has large amount of null values so impute them with statistical value.

It was found that each class owns certain age range hence class wise average age varies mostly, so filling null values with age mean corresponding to each class will be the effective step towards scoring maximum accurate analysis. This can be calculated using groupby(), fillna() and transform().

```
df.groupby('Pclass')['Age'].mean()
```

Pclass	
1	38.233441
2	29.877630
3	25.140620

```
Name: Age, dtype: float64
```

```
df['Age'].fillna(df.groupby('Pclass')['Age'].transform('mean')
)
```

Check for null values,

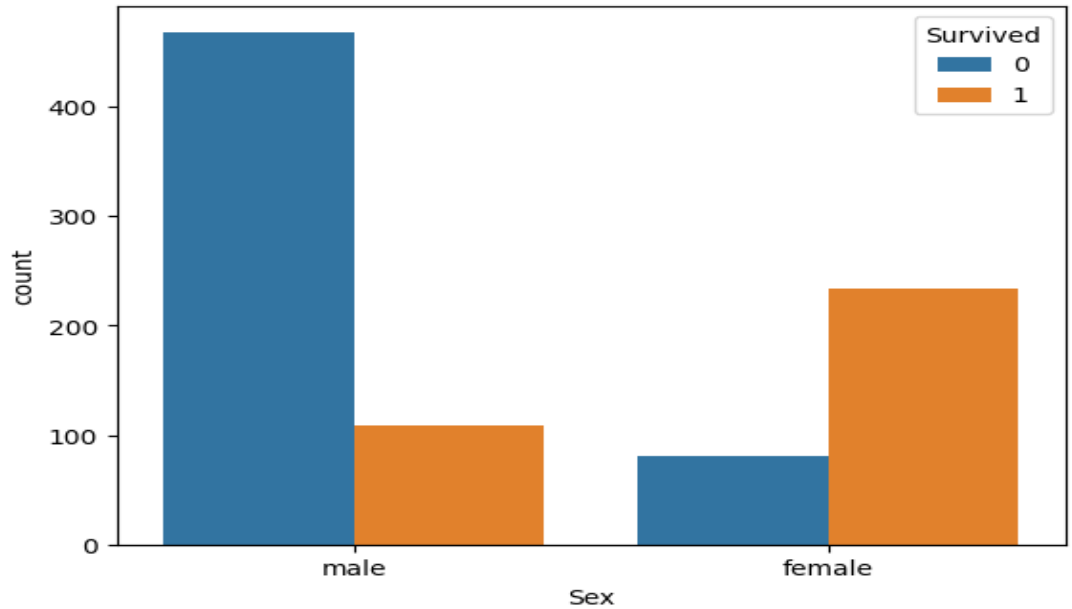
```
Survived     0

Pclass       0

Sex          0
```

```
Age          0
SibSp        0
Parch        0
dtype: int64
```

Visualizations:

1. Countplot:

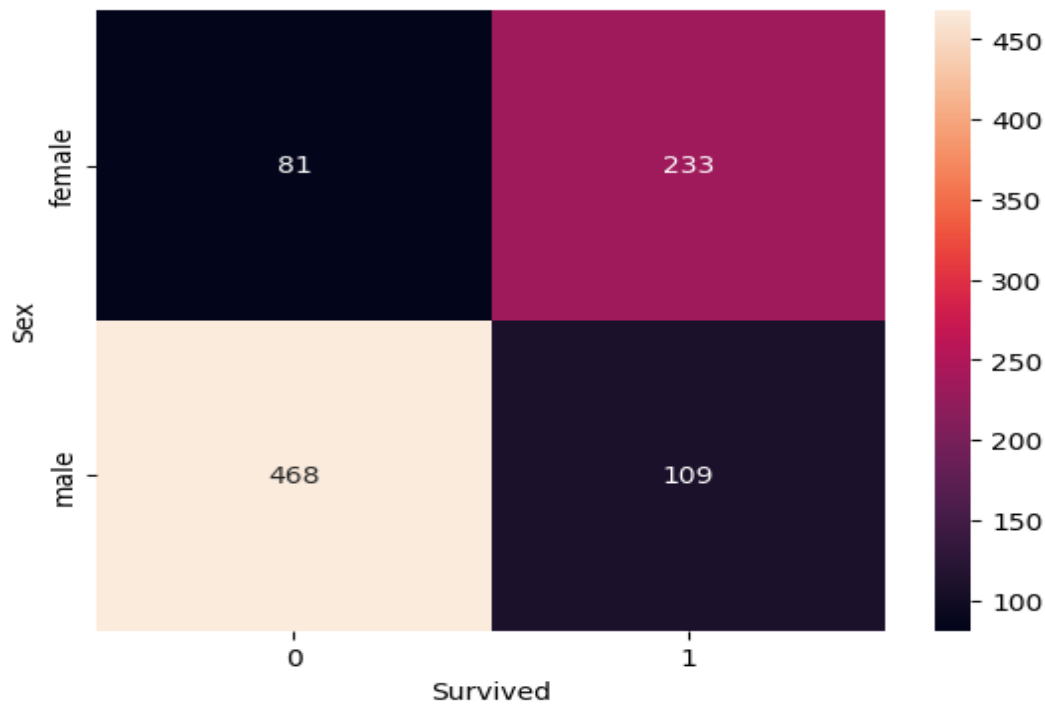
```
sns.countplot(data = df, x = 'Sex', hue = 'Survived')
plt.show()
```



Insight: Female passengers survived more than male.

2. Heatmaps:

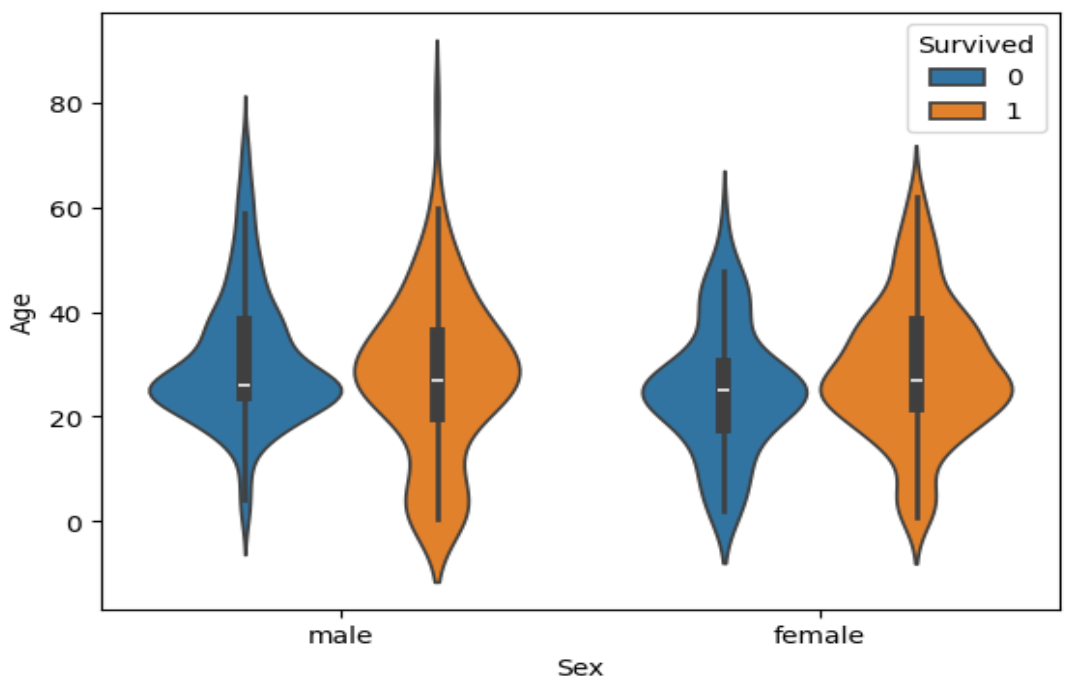
```
sns.heatmap(df.groupby(['Sex','Survived']).size().unstack()
,annot=True, fmt='d')
plt.show()
```



Insight: Heatmap exhibits light spectrum. Here, light shade shows most count whereas dark shows less count. Male has highest death count so it's in light shade.

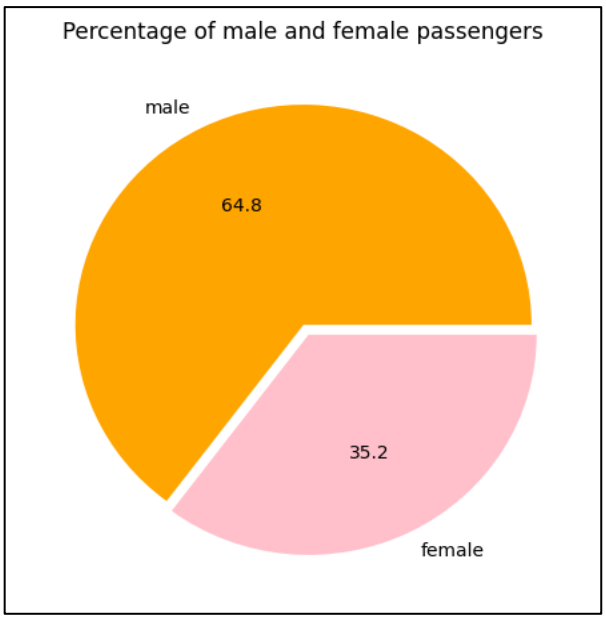
3. Violinplot:

```
sns.violinplot(data=df, x='Sex', y='Age', hue='Survived')
```



4. Pie Chart

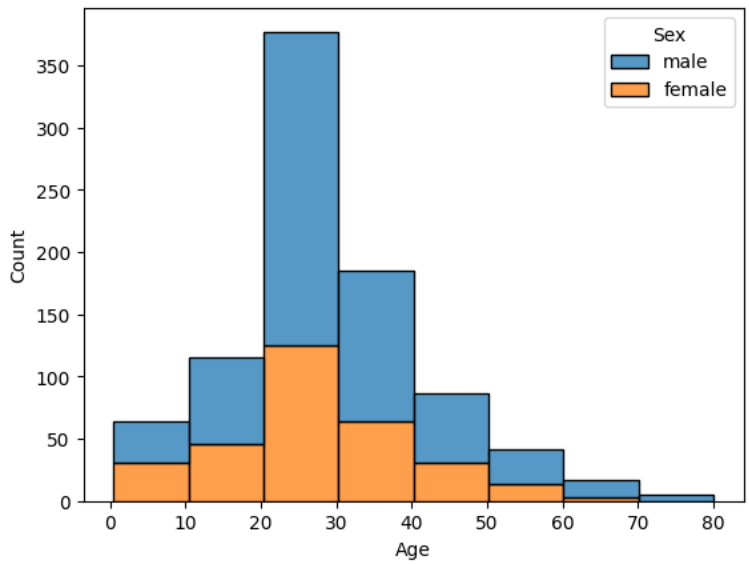
```
plt.pie(x=df['Sex'].value_counts(), labels=['male',
'female'], autopct="%0.1f", colors=['orange', 'pink'],
explode=[0, 0.05])
plt.title('Percentage of male and female passengers')
plt.tight_layout()
plt.show()
```



Insight : Male passengers are more in number than female.

5. Histplot

```
sns.histplot(data = df, x='Age', hue='Sex', bins=8,
multiple='stack')
plt.show()
```



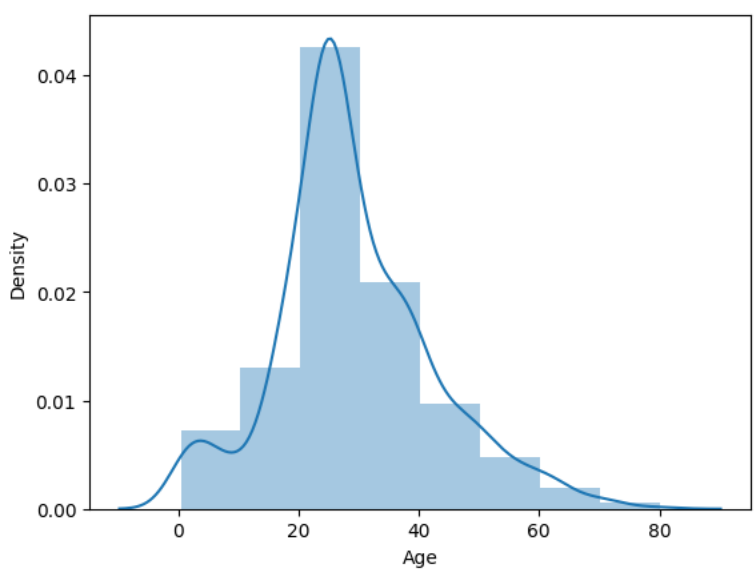
Insights:

- There are highest count of passengers in the 20-30 age range.
  - Within the range, males are in more number.
- The data is right skewed as the tail approaches towards higher ages.
- There are fewer old-aged passengers.

Skewness of data can be clearly illustrated with help of density plot.

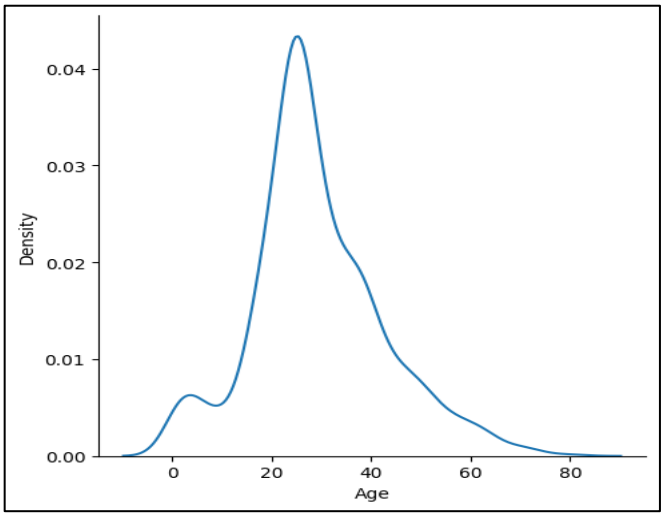
6. Distplot:

```
sns.distplot(df['Age'], bins=8)
```



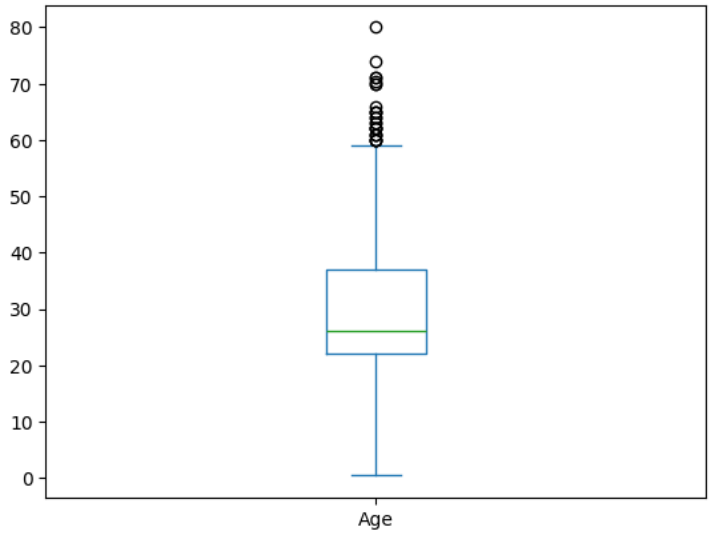
Similar type of graph can be achieved using displot.

```
sns.displot(df['Age'], kind='kde') (kde stands for density)
plt.show()
```



7. Boxplot: To find the outliers.

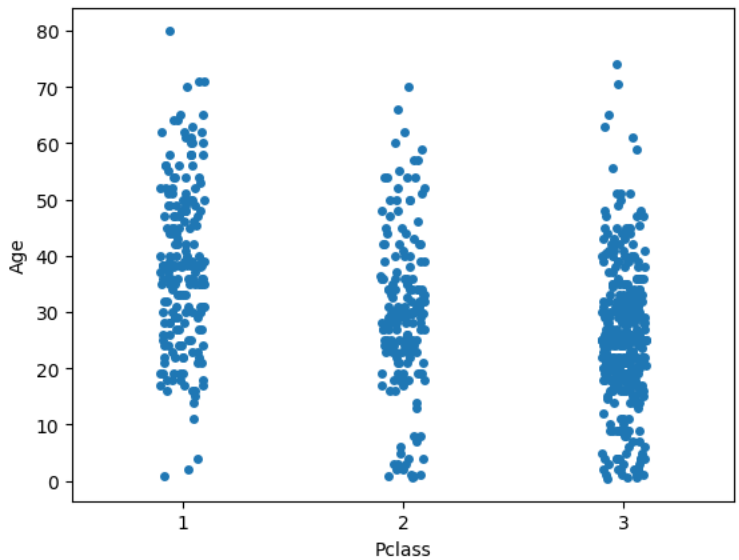
```
df['Age'].plot(kind= 'box')
plt.show()
```



Insight: There are outliers present above the maximum value.

8. Stripplot:

```
sns.stripplot(data=df, x= 'Pclass', y= 'Age', jitter=True)
plt.show()
```

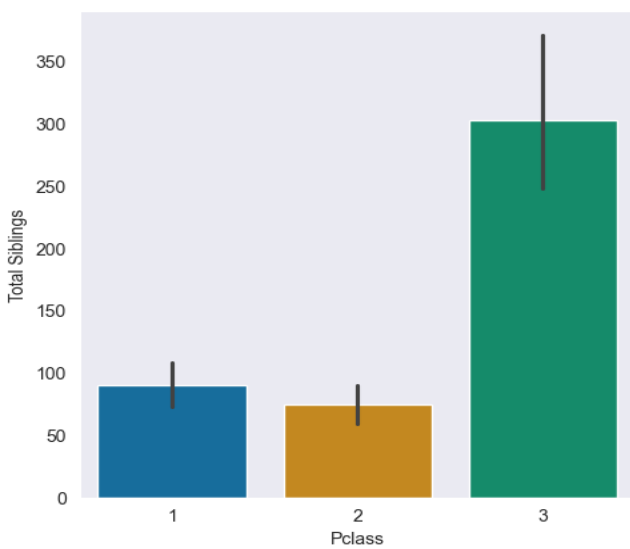


Insights:

- Pclass 3 is densely populated with young passengers.
- Pclass 1 has evenly spread passengers above 15 age.
- In Pclass 2, age is scatter above 35.

9. Catplot:

```
sns.set_style('dark')
sns.catplot(data=df, x='Pclass', y='SibSp', kind='bar',
estimator=np.sum, palette='colorblind')
plt.ylabel('Total Siblings')
plt.show()
```



Insight: Class 3 has highest siblings followed by class 1 and 2.